

CENTRAL BOARD OF SECONDARY EDUCATION
HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)
SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 4: DESCRIBING MOTION AROUND US

TIME ALLOWED: 45 MINUTES**MAXIMUM MARKS: 25****PAPER CODE: 086/CH04****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 A runner completes one full round of a circular track of radius 7 m. What is the ratio of the total distance covered to the magnitude of displacement?

- (A) 0
(B) 44 : 0 (Displacement is zero, distance is 44 m)
(C) 1 : 1
(D) 22 : 7

Q.2 Under which of the following conditions is the magnitude of average velocity equal to the average speed of an object?

- (A) When the object moves in a circular path
(B) When the object moves along a straight line in a single unchanging direction
(C) When the object travels back and forth
(D) Average velocity can never equal average speed

Q.3 A car accelerates uniformly from rest to a velocity of 72 km/h in 10 seconds. What is the acceleration of the car?

- (A) 7.2 m/s^2
(B) 2.0 m/s^2
(C) 4.0 m/s^2
(D) 20 m/s^2

Q.4 What does the slope of a distance-time graph represent?

- (A) Acceleration
(B) Speed
(C) Displacement
(D) Force

Q.5 What physical quantity does the area under a velocity-time graph represent?

- (A) Acceleration
(B) Distance or Displacement
(C) Speed
(D) Rate of change of velocity

Q.6 A body starts from rest and moves with uniform acceleration of 3 m/s^2 . What distance does it cover during the first 4 seconds?

- (A) 12 m
(B) 24 m
(C) 48 m
(D) 6 m

Q.7 Which equation of motion gives the relation between velocity and displacement without involving time explicitly?

- (A) $v = u + at$ (B) $s = ut + \frac{1}{2}at^2$
(C) $v^2 = u^2 + 2as$ (D) $s = \frac{u + v}{2} t$

Q.8 An object moves in a circular path of radius R with a constant speed v . Its acceleration is:

- (A) Zero because speed is constant (B) Non-zero and directed towards the centre of the circle
(C) Directed tangent to the circle (D) Constant in direction

Q.9 A train travelling at 20 m/s is brought to rest by applying brakes that produce a uniform retardation of 2 m/s^2 . The time taken to stop is:

- (A) 5 s (B) 10 s
(C) 20 s (D) 40 s

Q.10 The odometer of an automobile records:

- (A) Instantaneous speed (B) Average velocity
(C) Total distance travelled (D) Uniform acceleration

Q.11 A ball is thrown vertically upward with an initial velocity of 20 m/s. Taking $g = 10 \text{ m/s}^2$, what is the maximum height reached?

- (A) 10 m (B) 20 m
(C) 40 m (D) 200 m

Q.12 If the displacement-time graph of a moving object is a straight line parallel to the time axis, the object is:

- (A) Moving with uniform speed (B) Moving with uniform acceleration
(C) At rest (D) Moving with non-uniform velocity

Q.13 A particle moves with uniform acceleration. If its velocity changes from 10 m/s to 30 m/s while covering a distance of 80 m, the acceleration is:

- (A) 2.5 m/s^2 (B) 5.0 m/s^2
(C) 1.25 m/s^2 (D) 10 m/s^2

Q.14 Which of the following is a vector quantity?

- (A) Distance (B) Speed
(C) Displacement (D) Time

Q.15 A cyclist goes around a circular path of circumference 220 m in 44 seconds. What is the speed of the cyclist?

- (A) 5 m/s (B) 10 m/s
(C) 0.2 m/s (D) 50 m/s

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): The displacement of a moving body can be zero even if the distance travelled is non-zero.

Reason (R): Displacement is the vector representing the shortest path from initial to final position.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Motion in a circular track with constant speed is an accelerated motion.

Reason (R): Velocity is a vector, and in circular motion, the direction of motion changes continuously at every instant.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): The slope of a velocity-time graph gives the acceleration of the body.

Reason (R): Acceleration is defined as the time rate of change of displacement.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): An object can have constant speed and varying velocity.

Reason (R): Speed is a scalar quantity while velocity depends on both magnitude and direction.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): When a body moves with uniform velocity, its acceleration is non-zero.

Reason (R): Uniform velocity means velocity is changing at a uniform rate.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Velocity-Time Graph Analysis of an Electric Metro Train

An electric metro train starts from Station A and travels to Station B along a straight track. The velocity-time profile is recorded: During the first 10 seconds, it accelerates uniformly from 0 to 20 m/s. For the next 30 seconds, it maintains a constant cruise velocity of 20 m/s. Finally, brakes are applied and it decelerates uniformly to a complete stop in 10 seconds at Station B.

Q.21 What is the acceleration of the metro train during the first 10 seconds?

- (A) 1 m/s^2
 (B) 2 m/s^2
 (C) 4 m/s^2
 (D) 0.5 m/s^2

Q.22 What is the distance travelled by the train during the 30-second period of uniform velocity?

- (A) 300 m
 (B) 600 m
 (C) 450 m
 (D) 200 m

Q.23 What is the magnitude of retardation (deceleration) during the braking phase in the last 10 seconds?

(A) -2 m/s^2

(B) 2 m/s^2

(C) 1 m/s^2

(D) 0.5 m/s^2

Q.24 What is the total distance between Station A and Station B?

(A) 600 m

(B) 700 m

(C) 800 m

(D) 1000 m

Q.25 What was the average speed of the train for the entire journey from Station A to Station B?

(A) 16 m/s

(B) 20 m/s

(C) 12 m/s

(D) 10 m/s

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	Distance $s = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ m. Returning to the starting point gives displacement $= 0$ m.
Q.2	(B)	Along a straight line without reversing direction, distance equals displacement magnitude, so average speed equals average velocity.
Q.3	(B)	$u = 0$, $v = 72 \times \frac{5}{18} = 20$ m/s, $t = 10$ s. $a = \frac{v - u}{t} = \frac{20 - 0}{10} = 2.0$ m/s ² .
Q.4	(B)	Slope of distance-time graph $= \frac{\Delta \text{distance}}{\Delta \text{time}} = \text{Speed}$.
Q.5	(B)	Area under a velocity-time graph $= \text{velocity} \times \text{time} = \text{displacement}$ (or distance for uni-directional motion).
Q.6	(B)	$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(3)(4^2) = \frac{1}{2} \times 3 \times 16 = 24$ m.
Q.7	(C)	$v^2 - u^2 = 2as$ relates final velocity (v), initial velocity (u), acceleration (a), and displacement (s).
Q.8	(B)	In uniform circular motion, direction changes continuously; the resulting centripetal acceleration is directed radially inward.
Q.9	(B)	$u = 20$ m/s, $v = 0$, $a = -2$ m/s ² . $v = u + at \Rightarrow 0 = 20 - 2t \Rightarrow 2t = 20 \Rightarrow t = 10$ s.
Q.10	(C)	An odometer measures total distance covered in kilometers, whereas a speedometer measures instantaneous speed.
Q.11	(B)	At max height $v = 0$. $v^2 = u^2 - 2gh \Rightarrow 0 = 20^2 - 2(10)h \Rightarrow 20h = 400 \Rightarrow h = 20$ m.
Q.12	(C)	A horizontal line on a displacement-time graph means displacement does not change as time advances, so the object is stationary.
Q.13	(B)	$v^2 = u^2 + 2as \Rightarrow 30^2 = 10^2 + 2a(80) \Rightarrow 900 - 100 = 160a \Rightarrow 800 = 160a \Rightarrow a = 5$ m/s ² .
Q.14	(C)	Displacement has both magnitude and direction, making it a vector quantity.
Q.15	(A)	$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{220 \text{ m}}{44 \text{ s}} = 5$ m/s.
Q.16	(A)	If an object returns to its starting point, initial and final coordinates coincide, making displacement zero while distance is > 0 .
Q.17	(A)	Even though magnitude of velocity (speed) is fixed, change in direction means velocity changes, causing continuous centripetal acceleration.
Q.18	(C)	Assertion is true. Reason is false because acceleration is the rate of change of *velocity*, not displacement (rate of change of displacement is velocity).
Q.19	(A)	In uniform circular motion, speed is constant while direction changes, resulting in varying velocity.
Q.20	(D)	Both statements are false. Uniform velocity means velocity remains constant ($v - u = 0$), so acceleration is strictly zero.
Q.21	(B)	$a = \frac{20 - 0}{10} = 2$ m/s ² .
Q.22	(B)	$\text{Distance} = v \times t = 20 \text{ m/s} \times 30 \text{ s} = 600$ m.
Q.23	(B)	Retardation is the magnitude of negative acceleration: $a = \frac{0 - 20}{10} = -2$ m/s ² , so retardation $= 2$ m/s ² .
Q.24	(C)	Total distance is the area of trapezoid: $\text{Area} = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height} = \frac{1}{2} \times (50 + 30) \times 20 = 40 \times 20 = 800$ m.

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.25	(A)	$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{800 \text{ m}}{50 \text{ s}} = 16 \text{ m/s}$.